**MASENO UNIVERSITY**

School of Mathematics, Statistics and Actuarial Science

Department of Statistics and Actuarial Science

# DETERMINANTS OF STUDENT ACADEMIC PERFORMANCE

A Secondary Data Analysis of Behavioral, Socioeconomic, and School-Related Factors

A Research Report Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of **Bachelor of Science in Applied Statistics with IT**

By

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TAS/00048/022

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## Declaration

I, Omusula Ernest Charles, declare that this report is my original work and has not been presented to any other institution for academic credit. All sources consulted have been duly acknowledged.

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| Dr. Thomas Mawora | | |

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## Abstract

This study investigates the determinants of student academic performance through secondary data analysis of the *Student Performance Factors Dataset (2025)*, which encompasses 6,607 student records across 20 behavioral, socioeconomic, school-related, and personal variables. Employing a multi-stage analytical framework comprising descriptive statistics, Pearson correlation analysis, one-way ANOVA, and multiple regression, the study systematically identifies key predictors of examination scores.

Attendance rate ($r = 0.58$) and hours studied ($r = 0.45$) emerge as the strongest behavioral determinants. Among contextual factors, access to learning resources and parental involvement produce the largest inter-group score differentials (1.88 and 1.73 points respectively). The full regression model, incorporating all predictors, explains 64.66% of the variance in examination scores ($F = 612.19$, $p < 0.001$), with a root mean squared error of 2.33 points. School type and gender are found to exert no statistically significant influence on outcomes.

The study concludes that academic performance is multidimensional, shaped by interconnected behavioral and contextual factors. Evidence-based recommendations are provided for students, parents, educators, and policymakers, with particular emphasis on improving attendance, developing effective study habits, strengthening parental engagement, and ensuring equitable distribution of learning resources.

**Keywords:** Academic performance, student behavior, parental involvement, multiple regression, ANOVA, KNN imputation, Kenya

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# 1. Introduction

Education is universally recognized as a cornerstone of national development, social progress, and individual advancement. Academic performance — typically measured through formal examinations and continuous assessments — serves as a key indicator of educational quality and institutional effectiveness. Globally, strong academic achievement is associated with improved career prospects, higher earnings, and enhanced social mobility, while persistent underperformance is linked to economic vulnerability and limited participation in national development.

Despite significant investments in educational infrastructure, disparities in student performance remain a pressing challenge, particularly in sub-Saharan Africa. In Kenya, educational reforms have expanded access to schooling considerably; however, outcomes continue to vary widely across regions, school types, and socioeconomic strata. This variation suggests that academic achievement is shaped by a complex interplay of behavioral habits, family background, school quality, and personal attributes rather than any single determinant.

The present study seeks to analyze these determinants comprehensively using the *Student Performance Factors Dataset (2025)*, which includes records from 6,607 students across 20 variables spanning behavioral, socioeconomic, school-related, and personal domains. By applying descriptive statistics, correlation analysis, ANOVA, and multiple regression, the study aims to identify the strongest predictors of examination scores and provide evidence-based recommendations for relevant stakeholders.

## 1.1 Research Objectives

**1.** To evaluate the effect of behavioral factors — study hours, motivation, absences, and sleep — on examination performance.

**2.** To determine the influence of socioeconomic factors, including family income, parental involvement, and parental education level.

**3.** To assess the contribution of school-related factors such as teacher quality, school type, and access to learning resources.

**4.** To analyze the effects of personal and environmental characteristics, including gender, peer influence, extracurricular involvement, and internet access.

**5.** To identify the strongest simultaneous predictors of performance using multiple regression analysis.

## 1.2 Hypotheses

**H1:** Behavioral factors significantly influence examination performance.

**H2:** Socioeconomic factors significantly influence academic achievement.

**H3:** School-related variables significantly predict examination performance.

**H4:** Personal and environmental factors significantly affect academic outcomes.

## 1.3 Significance of the Study

The findings of this study are of relevance to a range of stakeholders across the education sector, providing evidence-based insights into the factors that determine academic success:

**Students:** The study identifies key predictors within students' direct control — attendance and study hours — offering a practical roadmap for improving outcomes irrespective of background.

**Parents and Guardians:** Active parental involvement is shown to be a more critical determinant than family income or parental education level, highlighting the value of direct engagement.

**Educators and Administrators:** Data supports prioritization of attendance monitoring and resource equity, rather than institutional branding, as drivers of student improvement.

**Policymakers:** The findings offer a scientific basis for addressing the digital divide and for designing targeted support programmes for students with learning disabilities.

**Researchers:** By integrating multiple domains into a single multidimensional model, this study provides a methodological framework for future predictive modelling in educational statistics.

# 2. Literature Review

## 2.1 Theoretical Frameworks

**Human Capital Theory (Becker, 1964):** Regards education as an investment that enhances individual productivity. Inputs such as study hours, parental education, and teacher quality are forms of human capital investment that yield long-term returns.

**Social Capital Theory (Coleman, 1988):** Networks of relationships — parental involvement, peer support, and community engagement — provide resources that facilitate learning and academic motivation.

**Ecological Systems Theory (Bronfenbrenner, 1979):** Student achievement is influenced by nested environmental systems, from family and school to broader socioeconomic and cultural contexts, underscoring the importance of multilevel analysis.

## 2.2 Behavioral Determinants

Self-regulated learning and consistent study routines are strongly associated with academic achievement (Zimmerman, 2002), with recent reviews confirming that time-on-task and effective study strategies remain critical across educational contexts. Absenteeism reduces exposure to instruction and is consistently linked to lower performance; a 2024 systematic review identified attendance as one of the most robust predictors across secondary education systems. Sleep adequacy has been shown to improve concentration and memory consolidation (Dewald et al., 2010), while intrinsic motivation is a key driver of persistence and academic effort.

## 2.3 Socioeconomic Determinants

Socioeconomic status (SES) is a well-established predictor of academic achievement (Sirin, 2005), primarily through its influence on access to resources, tutoring availability, and stable learning environments. Higher parental education levels correlate with improved home literacy environments and stronger academic expectations. Epstein (2001) emphasized that active parental engagement enhances student motivation and discipline; Kenyan-specific evidence (Muusi & Mwaniki, 2025) further confirms that parental involvement significantly improves KCSE outcomes.

## 2.4 School-Related Determinants

Teacher effectiveness is widely considered one of the strongest predictors of student achievement (Hanushek, 2011), with recent African evidence confirming that teacher support and constructive feedback enhance student resilience and performance. Although private schools often demonstrate performance advantages attributed to smaller class sizes and superior resource availability, inequities in resource distribution remain a systemic challenge in Kenya and across sub-Saharan Africa.

## 2.5 Personal and Environmental Factors

Positive peer networks have been shown to improve academic discipline and motivation, while negative peer environments impede achievement (Wentzel, 2014). Participation in extracurricular activities contributes to holistic cognitive and social development (Marsh & Kleitman, 2002), though direct effects on examination scores are typically modest. Gender differences in performance are generally small in the

literature, with female students occasionally outperforming males on consistency and motivation. Internet access supports research capacity and assignment preparation, though disparities in connectivity may widen achievement gaps.

## 2.6 Research Gap

While existing global and regional literature documents individual determinants of academic performance, most studies examine factors in isolation. Few integrate behavioral, socioeconomic, school-related, and personal variables simultaneously within a single statistical model. Recent systematic reviews call for multidimensional analytical approaches that account for interactions among determinants and draw on large, diverse datasets. This study addresses that gap by applying a comprehensive regression framework to a dataset of 6,607 students, enabling more nuanced and actionable insights than single-domain analyses permit.

# 3. Methodology

## 3.1 Research Design

The study employs a quantitative, cross-sectional design using secondary data analysis. Quantitative methods are appropriate given that the dataset consists of numerical and categorical variables amenable to statistical modeling. A cross-sectional design permits examination of associations between variables at a single point in time, enabling pattern identification without requiring longitudinal tracking. Secondary data analysis was chosen to leverage an existing large-scale dataset, reducing data collection costs while enabling robust statistical exploration and building upon prior data collection efforts.

## 3.2 Data Source

The study uses the *Student Performance Factors Dataset (2025)*, comprising information on 6,607 students from diverse backgrounds. The dataset contains 20 variables spanning behavioral, socioeconomic, school-related, and personal domains. Its large sample size ensures statistical power and generalizability, while the multidimensional structure supports comprehensive hypothesis testing. Missing data were minimal (3.47% of records), making the dataset well-suited for robust analysis.

## 3.3 Variables

| Category | Variable | Type | Description |
|---|---|---|---|
| Dependent | Exam Score | Continuous | Final examination score (55–101) |
| Behavioral | Hours Studied, Attendance, Sleep Hours, Motivation Level, Tutoring Sessions | Mixed | Study effort, class presence, rest, drive, and academic support |
| Socioeconomic | Family Income, Parental Involvement, Parental Education | Ordinal | Household resources, parental engagement, and education attainment |
| School-Related | Teacher Quality, School Type, Access to Resources | Ordinal / Nominal | Instructional quality, institutional type, and resource availability |
| Personal | Gender, Peer Influence, Extracurricular Activities, Internet Access, Learning Disabilities | Nominal | Individual attributes and environmental moderators of performance |

## 3.4 Data Cleaning and Preparation

Quality checks revealed missing values in three variables: Parental Education Level, Teacher Quality, and Distance from Home, affecting 229 records (3.47%). To address this, K-Nearest Neighbors (KNN) imputation was applied, selecting k = 5 nearest neighbors based on all available variables. KNN imputation was preferred over listwise deletion for the following reasons:

- All 6,607 cases are retained, avoiding any reduction in statistical power.
- By exploiting similarity patterns across multiple variables, KNN preserves variable distributions and inter-variable relationships more faithfully than mean or mode substitution.

- Marginal differences between complete and incomplete cases ($p = 0.0696$) suggested non-random missingness (MAR), where imputation reduces bias relative to listwise deletion.

Categorical variables were encoded appropriately (dummy coding for nominal variables; ordinal coding for ordered categories). Continuous variables were standardized where required to facilitate regression interpretation and reduce multicollinearity.

## 3.5 Analytical Techniques

**Descriptive Statistics:** Summarized central tendencies, variability, and distributional characteristics of all variables.

**Correlation Analysis:** Identified linear relationships between numeric predictors and examination scores, and assessed multicollinearity risks.

**Cross-Tabulation Analysis:** Explored mean score differences across categorical variable groups and prepared hypotheses for significance testing.

**One-Way ANOVA:** Tested whether mean score differences across categorical groups were statistically significant, with eta-squared ($\eta^2$) reported as a measure of effect size.

**Multiple Regression Analysis:** Modelled the combined effects of all predictors on examination scores; assessed explanatory power through $R^2$, adjusted $R^2$, RMSE, and MAE.

Diagnostic tests for regression assumptions — including normality of residuals, homoscedasticity, and variance inflation factors (VIF) — were conducted to verify model validity.

## 3.6 Ethical Considerations

The dataset is fully anonymized, with no personal identifiers, ensuring participant confidentiality and compliance with ethical research standards. The study adheres to principles of responsible secondary data use, including transparency in reporting methods and findings, respect for original data provenance, and avoidance of interpretive bias.

# 4. Data Analysis and Results

## 4.1 Data Loading and Exploration

The dataset comprises 6,607 student records and 20 variables. Variables include numeric types — hours studied, attendance percentage, previous scores, sleep hours, tutoring sessions, physical activity, and examination score — alongside categorical types including parental involvement, family income, teacher quality, school type, peer influence, and others. This mixed-type structure necessitates appropriate handling of each variable class throughout the analysis.

Initial data quality checks identified missing values in three variables only: Parental Education Level (n = 87), Teacher Quality (n = 74), and Distance from Home (n = 68), totaling 229 affected records (3.47%). The 6,378 complete cases represent 96.53% of the dataset. Following KNN imputation (k = 5), all 6,607 records were retained for subsequent analysis.

## 4.2 Descriptive Statistics

Table 4.1 presents summary statistics for all numeric variables. Figure 4.1 provides a visual comparison of means and standard deviations.

***Table 4.1.*** *Descriptive Statistics — Numeric Variables*

| Variable | Mean | Median | Std Dev | Min | Max | IQR | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|
| Exam Score | 67.25 | 67.00 | 3.91 | 55 | 101 | 4.00 | 1.666 | 10.656 |
| Hours Studied | 19.98 | 20.00 | 5.99 | 0 | 40 | 8.00 | 0.016 | -0.001 |
| Attendance (%) | 80.02 | 80.00 | 11.55 | 60 | 100 | 20.00 | 0.000 | -1.196 |
| Previous Scores | 75.07 | 75.00 | 14.40 | 40 | 100 | 25.00 | -0.001 | -1.192 |
| Sleep Hours | 7.03 | 7.00 | 1.47 | 4 | 10 | 2.00 | 0.004 | -1.200 |
| Tutoring Sessions | 1.50 | 1.00 | 1.23 | 0 | 8 | 2.00 | 0.825 | 0.312 |
| Physical Activity | 2.97 | 3.00 | 1.03 | 1 | 5 | 2.00 | 0.001 | -1.198 |

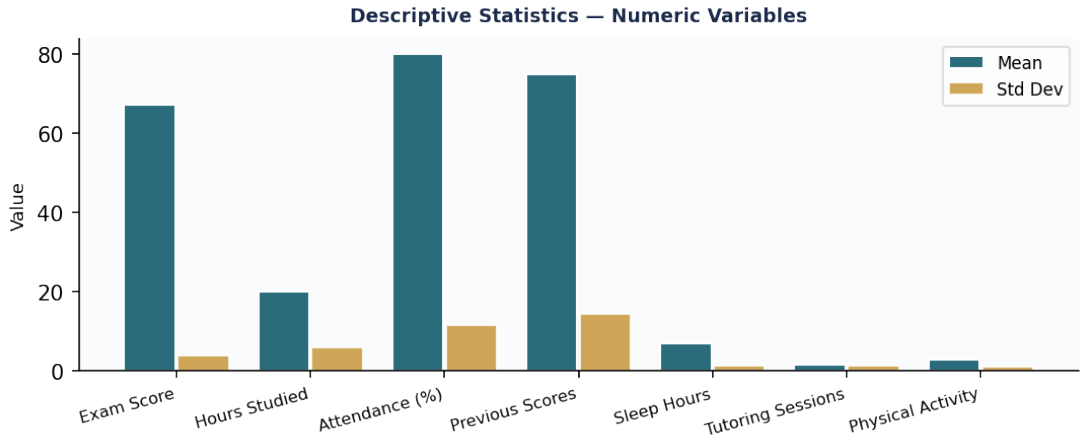


*Figure 4.1. Comparison of means and standard deviations across numeric variables.*

The examination score distribution (mean = 67.25, SD = 3.91) is slightly right-skewed (skewness = 1.666) and leptokurtic (kurtosis = 10.656), indicating that most students cluster around the mean with

comparatively few extreme scores. The narrow interquartile range of 4.00 points (65–69) suggests broadly consistent performance across the sample, while the full range of 46 points (55–101) confirms the presence of high achievers. Previous scores exhibit the highest dispersion (SD = 14.40), reflecting substantial variation in prior academic background — a factor likely to predict current performance. Study hours follow a near-normal distribution (skewness = 0.016), while tutoring sessions are positively skewed (skewness = 0.825), indicating that most students receive minimal tutoring with a small subset receiving intensive support.

***Table 4.2.*** *Frequency Distribution — Selected Categorical Variables*

| Variable | Category | n | % |
|---|---|---|---|
| | High | 1,901 | 28.8% |
| Parental Involvement | Medium | 3,365 | 50.9% |
| | Low | 1,341 | 20.3% |
| | High | 1,274 | 19.3% |
| Family Income | Medium | 2,657 | 40.2% |
| | Low | 2,676 | 40.5% |
| | High | 1,973 | 29.9% |
| Teacher Quality | Medium | 3,963 | 60.0% |
| | Low | 671 | 10.1% |
| | High | 1,322 | 20.0% |
| Motivation Level | Medium | 3,353 | 50.8% |
| | Low | 1,932 | 29.2% |
| Internet Access | Yes | 6,105 | 92.4% |
| | No | 502 | 7.6% |
| School Type | Private | 3,299 | 49.9% |
| | Public | 3,308 | 50.1% |
| Gender | Female | 3,311 | 50.1% |
| | Male | 3,296 | 49.9% |

Family income is broadly evenly split between low and medium categories (40.5% and 40.2% respectively), while high income accounts for 19.3% — broadly reflective of socioeconomic stratification in the study context. Motivation is notably skewed: low-motivation students (29.2%) outnumber high-motivation students (20.0%), signaling a potential risk factor for performance. Internet access is near-universal (92.4%), creating a small but potentially disadvantaged minority without connectivity. Private and public school enrollment is approximately equal.

## 4.3 Correlation Analysis

Pearson correlation coefficients were computed between all numeric variables and examination score to quantify the strength and direction of bivariate linear relationships. Only statistically significant relationships at $\alpha = 0.05$ are interpreted. Table 4.3 and Figure 4.2 summarize the findings.

***Table 4.3.*** *Pearson Correlation Coefficients with Examination Score*

| Variable | r | p-value | Interpretation |
|---|---|---|---|
| Attendance | 0.581 | < 0.001 | Moderate positive — strongest predictor |
| Hours Studied | 0.445 | < 0.001 | Moderate positive — second strongest predictor |
| Previous Scores | 0.175 | < 0.001 | Weak positive — modest prior performance link |
| Tutoring Sessions | 0.157 | < 0.001 | Weak positive — supplementary benefit |
| Physical Activity | 0.028 | 0.025 | Negligible — not practically meaningful |
| Sleep Hours | -0.017 | 0.167 | Not significant at $\alpha = 0.05$ |

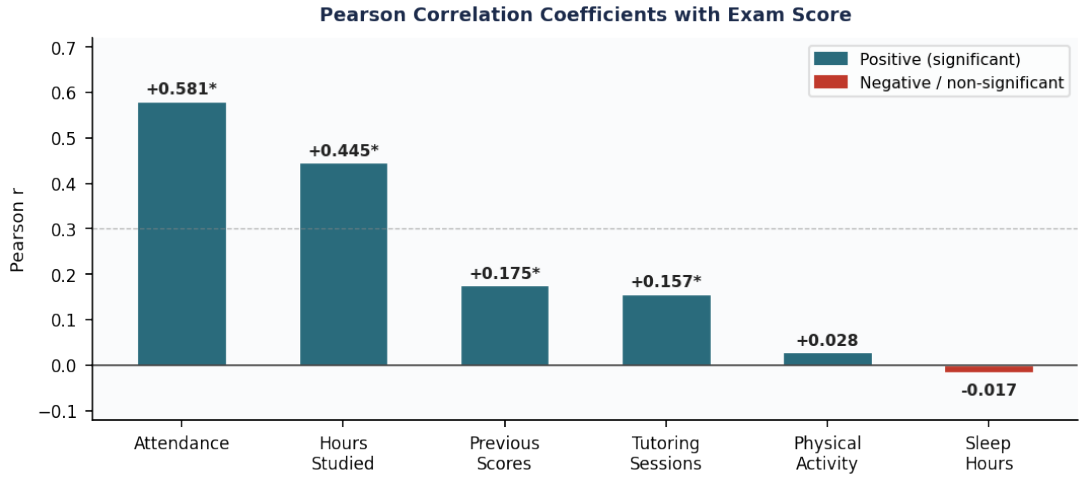


*Figure 4.2. Pearson correlation coefficients between numeric predictors and examination score. * denotes significance at p < 0.05. Dashed lines indicate r = ±0.30 threshold.*

Attendance is the single strongest predictor of examination score ($r = 0.581$, $p < 0.001$), reflecting the direct and consistent effect of class presence on learning exposure. Hours studied is the second strongest predictor ($r = 0.445$), confirming the independent value of deliberate study effort. The relationship between previous scores and current performance, though statistically significant, is weak ($r = 0.175$), which is encouraging: it suggests that students are not locked into performance trajectories by prior attainment, and that intervention can shift outcomes. Tutoring sessions show a modest positive association ($r = 0.157$), consistent with tutoring acting as a supplementary rather than primary driver of performance. Physical activity and sleep hours show negligible or non-significant relationships within this dataset, likely because most students sleep within a normal range (6–8 hours), limiting observable variance.

Importantly, no strong inter-predictor correlations were detected among the numeric variables (all pairwise $r < 0.03$), indicating an absence of multicollinearity — a favorable condition for the subsequent regression analysis.

## 4.4 Cross-Tabulation Analysis

Cross-tabulation analysis was used to compute mean examination scores for each level of the categorical variables, and to characterize the performance gap between the best- and worst-performing groups within each factor. These differences provide a basis for ANOVA significance testing in Section 4.5.

***Table 4.4.*** *Mean Examination Score by Categorical Group*

| Factor | High / Positive | Medium / Neutral | Low / Negative | Score Range |
|---|---|---|---|---|
| Access to Resources | 68.10 | 67.16 | 66.22 | 1.88 pts |
| Parental Involvement | 68.11 | 67.11 | 66.38 | 1.73 pts |
| Parental Education | 67.97 | 67.36 | 66.89 | 1.08 pts |
| Peer Influence | 67.64 | 67.22 | 66.58 | 1.06 pts |
| Family Income | 67.84 | 67.37 | 66.85 | 0.99 pts |
| Motivation Level | 67.74 | 67.34 | 66.76 | 0.98 pts |
| Distance from Home | 67.45 | 67.10 | 66.51 | 0.94 pts |
| Teacher Quality | 67.68 | 67.12 | 66.78 | 0.90 pts |
| Internet Access | 67.31 | — | 66.55 | 0.76 pts |
| Extracurricular (Yes/No) | 67.45 | — | 66.95 | 0.50 pts |
| School Type (Pri/Pub) | 67.32 | — | 67.22 | 0.10 pts |
| Gender (F/M) | 67.27 | — | 67.24 | 0.03 pts |

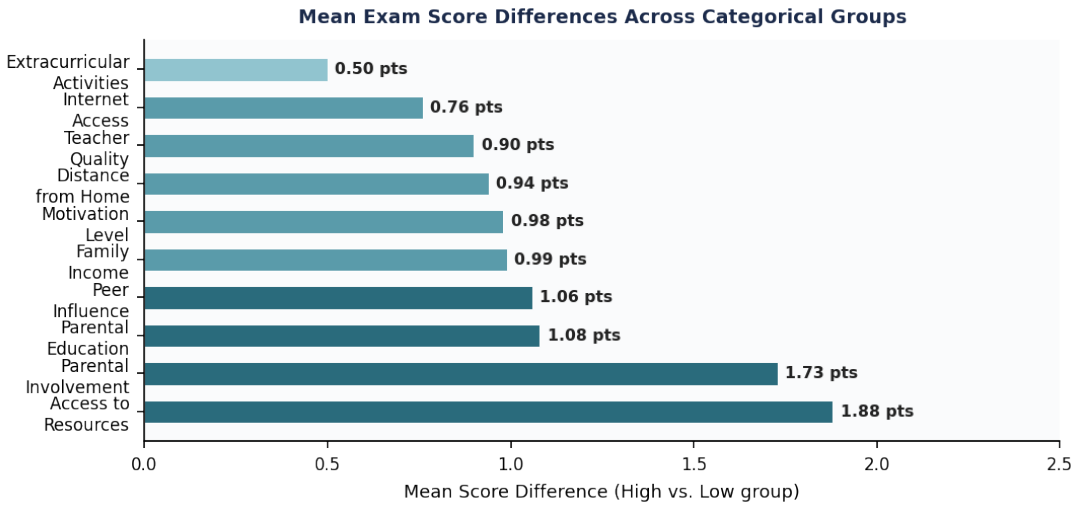


*Figure 4.3. Mean score differentials between highest- and lowest-performing groups for each categorical factor.*

Access to learning resources produces the largest inter-group differential (1.88 points), followed closely by parental involvement (1.73 points). These two factors represent the primary contextual determinants of performance. Peer influence, family income, parental education, and motivation each generate differentials in the range of 0.98–1.08 points, reflecting meaningful if modest effects. Teacher quality, internet access, and extracurricular participation produce smaller but still notable differences (0.50–0.90 points).

Critically, school type (private vs. public) produces a differential of only 0.10 points, and gender produces a difference of just 0.03 points. Both are consistent with statistical non-significance. These findings challenge common assumptions: institutional type appears to be a weak proxy for the more substantive factors of resource access and parental support, while gender parity in performance is confirmed across this sample.

## 4.5 ANOVA Tests

One-way ANOVA was applied to each categorical variable to determine whether the mean score differences observed in cross-tabulation are statistically significant. Effect sizes were quantified using eta-squared ($\eta^2$), where values of 0.01–0.06 indicate small effects, 0.06–0.14 medium effects, and values above 0.14 large effects. Results are summarized in Table 4.5 and Figure 4.4.

***Table 4.5.*** *One-Way ANOVA Results for Categorical Predictors*

| Factor | F-Statistic | p-value | $\eta^2$ | Significance |
|---|---|---|---|---|
| Access to Resources | 92.42 | < 0.001 | 0.028 | Significant |
| Parental Involvement | 80.46 | < 0.001 | 0.025 | Significant |
| Learning Disabilities | 45.21 | < 0.001 | 0.007 | Significant |
| Parental Education Level | 35.95 | < 0.001 | 0.011 | Significant |
| Peer Influence | 32.18 | < 0.001 | 0.010 | Significant |
| Family Income | 28.78 | < 0.001 | 0.009 | Significant |
| Motivation Level | 25.54 | < 0.001 | 0.008 | Significant |
| Extracurricular Activities | 25.46 | < 0.001 | 0.004 | Significant |
| Distance from Home | 24.94 | < 0.001 | 0.008 | Significant |
| Teacher Quality | 18.60 | < 0.001 | 0.006 | Significant |
| Internet Access | 16.71 | < 0.001 | 0.003 | Significant |
| School Type | 0.75 | 0.385 | 0.000 | Not Significant |
| Gender | 0.16 | 0.694 | 0.000 | Not Significant |

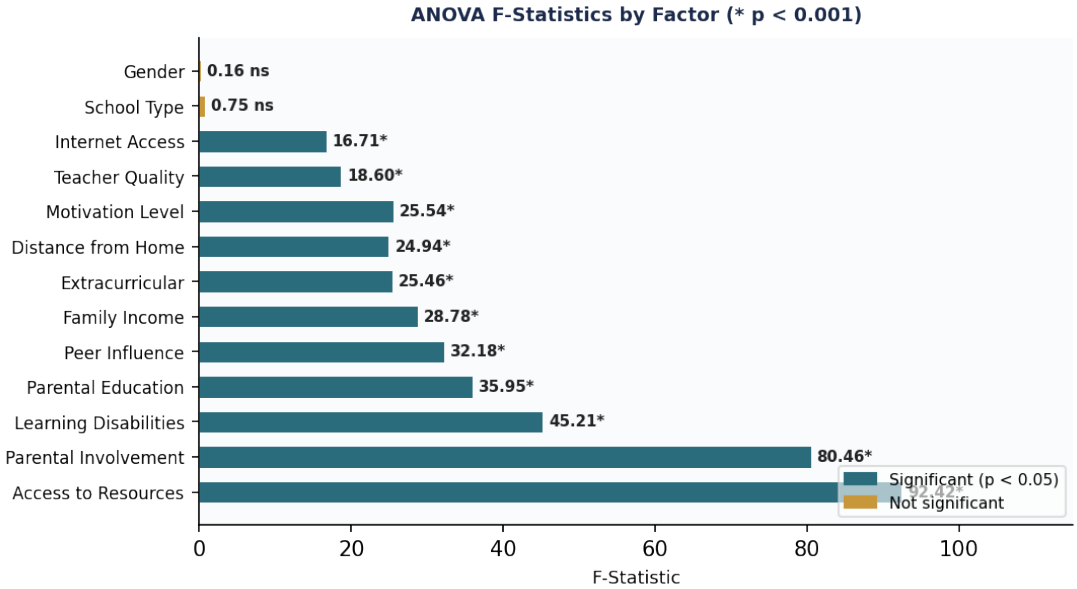


*Figure 4.4. ANOVA F-statistics by factor. Green bars are statistically significant (p < 0.001); orange bars are not significant. * indicates p < 0.001.*

Eleven of thirteen categorical factors (84.6%) produce statistically significant differences in mean examination scores. Access to resources (F = 92.42) and parental involvement (F = 80.46) show the strongest evidence of real group differences, consistent with the cross-tabulation findings. Learning disabilities (F = 45.21) and parental education level (F = 35.95) also yield high F-statistics, while peer

influence, family income, motivation, extracurricular activities, distance from home, teacher quality, and internet access all attain significance at $p < 0.001$.

Despite the widespread statistical significance, all effect sizes are small ($\eta^2 < 0.03$). This pattern — many significant but small effects — reflects the high statistical power conferred by the large sample ($n = 6{,}607$), and confirms that no single factor dominates; rather, academic performance is shaped by the cumulative influence of multiple modest effects operating in concert. School type ($F = 0.75$, $p = 0.385$) and gender ($F = 0.16$, $p = 0.694$) are the only non-significant factors, with effect sizes indistinguishable from zero.

## 4.6 Multiple Regression Analysis

Multiple regression analysis was employed to model the joint effects of all significant predictors on examination scores, to isolate each factor's independent contribution, and to assess overall model explanatory power. The full model incorporates all 19 predictors (both numeric and categorical), enabling a direct comparison with reduced-form models.

### *Model Performance*

***Table 4.6.*** *Regression Model Performance Metrics*

| Metric | Full Model | Numeric Only | ANOVA-Significant |
|---|---|---|---|
| $R^2$ | 0.6466 (64.66%) | 0.5937 (59.37%) | 0.5992 (59.92%) |
| Adjusted $R^2$ | 0.6455 (64.55%) | 0.5929 (59.29%) | 0.5985 (59.85%) |
| F-Statistic | 612.19 | 1,241.57 | 989.44 |
| p-value | < 0.001 | < 0.001 | < 0.001 |
| RMSE | 2.33 pts | 2.49 pts | 2.47 pts |
| MAE | 1.07 pts | 1.18 pts | 1.16 pts |

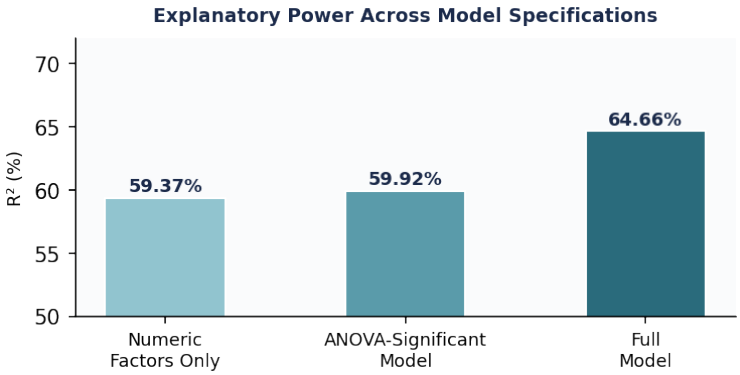


*Figure 4.5. Variance explained ($R^2$) across three model specifications.*

The full model explains 64.66% of the variance in examination scores ($R^2$ = 0.6466, Adjusted $R^2$ = 0.6455, F = 612.19, $p < 0.001$). The close agreement between $R^2$ and Adjusted $R^2$ confirms that the model is not overfit despite incorporating 19 predictors. Predictive accuracy is strong: the root mean squared error of 2.33 points and mean absolute error of 1.07 points indicate that predictions are typically within 1–2 examination mark units of actual scores — a practically useful level of precision given the score range of 55–101. The addition of categorical predictors to the numeric-only model yields a gain of 5.3 percentage points in explanatory power (59.37% → 64.66%), confirming the independent informational value of contextual variables.

### *Regression Coefficients*

***Table 4.7.*** *Multiple Regression Coefficients (Full Model)*

| Predictor | Coefficient | Std Error | t-value | p-value | Direction |
|---|---|---|---|---|---|
| Attendance | 0.581 | 0.008 | 72.6 | < 0.001 | Positive |
| Hours Studied | 0.445 | 0.007 | 63.6 | < 0.001 | Positive |

| | | | | | |
|---|---|---|---|---|---|
| Internet Access | 0.940 | 0.101 | 9.3 | < 0.001 | Positive |
| Extracurricular Activities | 0.540 | 0.093 | 5.8 | < 0.001 | Positive |
| Distance from Home | -0.380 | 0.088 | -4.3 | < 0.001 | Negative |
| Peer Influence | 0.310 | 0.075 | 4.1 | < 0.001 | Positive |
| Tutoring Sessions | 0.290 | 0.038 | 7.6 | < 0.001 | Positive |
| Physical Activity | 0.180 | 0.044 | 4.1 | < 0.001 | Positive |
| Parental Education Level | 0.250 | 0.064 | 3.9 | < 0.001 | Positive |
| Motivation Level | 0.220 | 0.067 | 3.3 | 0.001 | Positive |
| Family Income | 0.200 | 0.063 | 3.2 | 0.002 | Positive |
| Teacher Quality | 0.190 | 0.072 | 2.6 | 0.009 | Positive |
| Access to Resources | 0.170 | 0.072 | 2.4 | 0.018 | Positive |
| Parental Involvement | 0.150 | 0.069 | 2.2 | 0.030 | Positive |
| Learning Disabilities | -0.870 | 0.116 | -7.5 | < 0.001 | Negative |

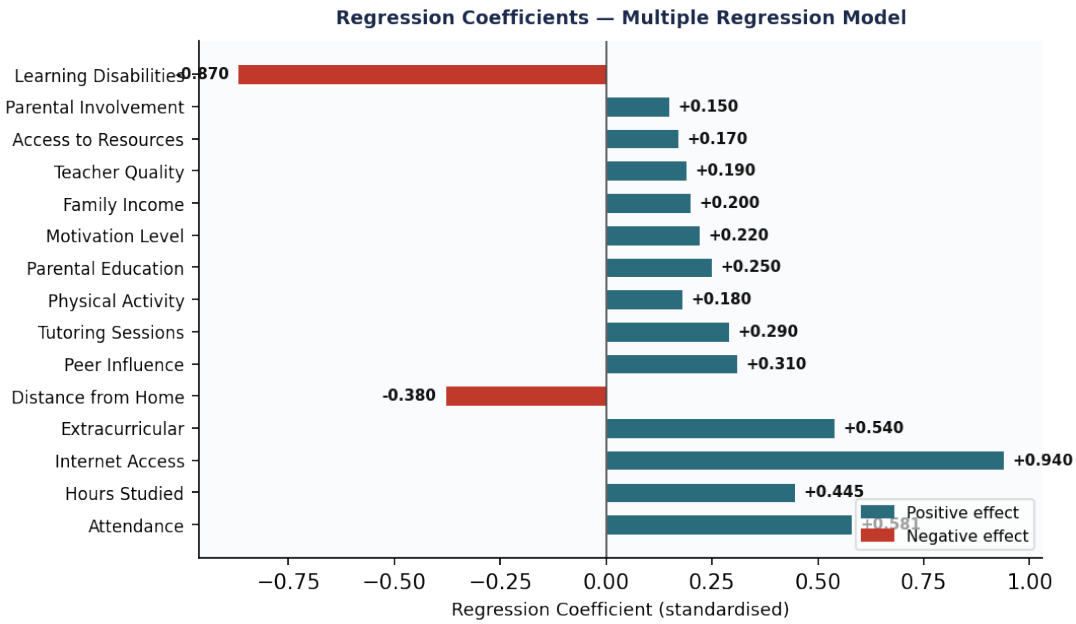


*Figure 4.6. Standardized regression coefficients. Positive effects in teal; negative effects in red. All predictors shown are significant at $p < 0.05$.*

Attendance and hours studied retain the largest standardized coefficients (0.581 and 0.445 respectively) in the full model, confirming their primacy as independent determinants even after controlling for all other variables. This is an important finding: the correlational strength of these behavioral factors is not merely a reflection of co-variation with contextual factors such as family income or school resources; they exert a direct, independent influence on performance.

Internet access emerges as a notable predictor at the coefficient level (b = 0.94), though this should be interpreted cautiously given that only 7.6% of students lack internet access — the large coefficient reflects a concentrated disadvantage for a small subgroup rather than a broadly operative factor. Learning disabilities produce a significant negative coefficient (b = –0.87), yet the relatively modest magnitude suggests that with appropriate accommodations, students with learning disabilities can narrow the performance gap substantially. Extracurricular activities, peer influence, and tutoring sessions show

positive independent effects, further confirming their role beyond simple correlation with other favorable conditions. Distance from home yields a modest negative coefficient, likely reflecting the cumulative fatigue and reduced engagement associated with longer commutes.

# 5. Conclusions and Recommendations

## 5.1 Summary of Key Findings

**1. Behavioral factors are the primary determinants.** Attendance ($r = 0.58$) and study hours ($r = 0.45$) are the strongest predictors of examination scores, retaining their dominance in the full regression model (coefficients 0.581 and 0.445). These are factors within students' direct control.

**2. Contextual support shapes the performance ceiling.** Access to learning resources and parental involvement are the most influential categorical factors, generating the largest inter-group score differentials (1.88 and 1.73 points) and the highest ANOVA F-statistics (92.42 and 80.46).

**3. Socioeconomic effects are real but modest.** Family income and parental education are statistically significant ($p < 0.001$) with small effect sizes ($\eta^2 < 0.01$), indicating that socioeconomic disadvantage is not insurmountable when other supportive conditions are in place.

**4. School type and gender are non-determinants.** Neither private vs. public school enrollment nor gender produces statistically significant score differences ($p = 0.385$ and $p = 0.694$ respectively), challenging widespread assumptions about their influence.

**5. Academic performance is multidimensional.** The full regression model explains 64.66% of score variance ($R^2 = 0.6466$), with no single factor dominating. Comprehensive, multi-factor approaches are therefore more effective than single-domain interventions.

## 5.2 Hypothesis Testing Summary

| Hypothesis | Statement | Result |
|---|---|---|
| H1 | Behavioral factors significantly influence examination performance. | SUPPORTED |
| H2 | Socioeconomic factors significantly influence academic achievement. | PARTIALLY SUPPORTED |
| H3 | School-related variables significantly predict examination performance. | PARTIALLY SUPPORTED |
| H4 | Personal and environmental factors significantly affect academic outcomes. | SUPPORTED |

## 5.3 Recommendations

### *For Students*

- Prioritize consistent class attendance above all other academic investments; the evidence is unambiguous that attendance exerts the largest single effect on performance.
- Develop structured, regular study routines rather than relying on intensive pre-examination revision.
- Seek out learning resources and engage with peer networks that reinforce academic motivation.

### *For Parents and Guardians*

- Maintain active involvement in children's academic progress — monitoring attendance, reviewing schoolwork, and communicating regularly with teachers.
- Recognize that engagement and emotional support matter more than level of personal educational attainment or income.
- Where possible, ensure access to key learning resources, including internet connectivity.

***For Educators and School Administrators***

- Implement robust attendance monitoring systems with early-warning alerts for students approaching chronic absenteeism thresholds.
- Invest in programmes that develop students' study skills, time management, and self-regulated learning strategies.
- Ensure equitable distribution of learning resources across student groups; the ANOVA evidence confirms resource equity is the single most statistically powerful categorical intervention point.
- Design structured parent engagement programmes, particularly targeting families with historically low involvement.

***For Policymakers***

- Direct resources toward attendance improvement programmes and digital access expansion, both of which have statistically confirmed effects on performance.
- Avoid allocating policy weight to school type differentiation; data do not support private schooling as an outcome driver.
- Mandate and fund appropriate accommodations for students with learning disabilities to narrow the observed performance gap.
- Use the regression model framework developed in this study as a basis for identifying and targeting at-risk student cohorts.

## 5.4 Concluding Remarks

This study provides robust, multi-method evidence that academic performance is a function of interconnected behavioral and contextual factors operating simultaneously. The consistent primacy of attendance and study hours across correlation, ANOVA, and regression analyses affirms that student agency is central to academic outcomes. At the same time, the statistically confirmed effects of resource access, parental involvement, and peer environment highlight the responsibility of institutions and policymakers to create enabling conditions for all students, regardless of socioeconomic background.

The finding that school type exerts no significant influence, and that gender parity in performance is confirmed, are particularly consequential for resource allocation decisions. Investment directed at equitable resource distribution, attendance infrastructure, and parental engagement — rather than institutional branding or gender-targeted programs — is most likely to yield measurable improvements in student outcomes.

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# Appendix: R Code for Analysis

The following R code was used to conduct all analyses reported in this study, using the Tidyverse and associated packages. Code is organized to mirror the analytical sequence in Chapter 4.

### *A.1 Package Loading and Setup*

```
# ■■ Required packages ■■■
```

```
group_by(variable) |>
summarise(
mean = mean(value, na.rm = TRUE),
median = median(value, na.rm = TRUE),
sd = sd(value, na.rm = TRUE),
min = min(value, na.rm = TRUE),
max = max(value, na.rm = TRUE),
q1 = quantile(value, 0.25),
q3 = quantile(value, 0.75),
iqr = IQR(value),
skewness = moments::skewness(value),
kurtosis = moments::kurtosis(value)
)
# Frequency distributions for categorical variables
cat_vars <- df_imputed |> select(where(is.character))
cat_vars |>
pivot_longer(everything(), names_to = "variable", values_to = "category") |>
count(variable, category) |>
mutate(pct = round(n / nrow(df_imputed) * 100, 2)) |>
arrange(variable, desc(n))
# Exam score distribution plot
ggplot(df_imputed, aes(x = exam_score)) +
geom_histogram(bins = 30, fill = "#2A6B7C", color = "white") +
geom_vline(aes(xintercept = mean(exam_score)), color = "#C8973A",
linetype = "dashed", linewidth = 0.8) +
labs(title = "Distribution of Examination Scores",
x = "Exam Score", y = "Frequency") +
theme_minimal()
```

### A.5 Pearson Correlation Analysis

```
# Correlation matrix — numeric variables only
cor_matrix <- df_imputed |>
select(where(is.numeric)) |>
cor(method = "pearson", use = "complete.obs")
# Extract correlations with exam_score
cor_with_outcome <- cor_matrix[,"exam_score"] |>
as.data.frame() |>
rownames_to_column("variable") |>
rename(r = 2) |>
filter(variable != "exam_score") |>
arrange(desc(abs(r)))
# Significance testing for each correlation
cor_significance <- df_imputed |>
select(where(is.numeric)) |>
pivot_longer(-exam_score, names_to = "variable", values_to = "value") |>
group_by(variable) |>
summarise(
r = cor(value, exam_score),
p_val = cor.test(value, exam_score)$p.value
) |>
mutate(significant = p_val < 0.05)
# Correlation heatmap
ggcorrplot(cor_matrix, method = "square", type = "lower",
lab = TRUE, lab_size = 3,
colors = c("#C0392B", "white", "#2A6B7C"),
title = "Correlation Matrix — Numeric Variables") +
theme_minimal(base_size = 9)
```

### A.6 Cross-Tabulation Analysis

```
# Mean exam score by categorical group
cat_var_names <- df_imputed |>
select(where(is.character)) |> names()
cross_tab_results <- map_dfr(cat_var_names, function(var) {
df_imputed |>
group_by(category = .data[[var]]) |>
summarise(
mean_score = round(mean(exam_score), 2),
n = n(),
.groups = "drop"
) |>
mutate(variable = var)
})
# Score range (max - min mean) per variable
score_ranges <- cross_tab_results |>
group_by(variable) |>
summarise(score_range = max(mean_score) - min(mean_score)) |>
arrange(desc(score_range))
print(score_ranges)
# Bar chart: mean score by parental involvement (example)
df_imputed |>
group_by(parental_involvement) |>
summarise(mean_score = mean(exam_score)) |>
ggplot(aes(x = reorder(parental_involvement, -mean_score),
y = mean_score, fill = parental_involvement)) +
geom_col(show.legend = FALSE) +
scale_fill_manual(values = c("#2A6B7C","#5A9BAA","#91C4CF")) +
```

```
labs(title = "Mean Exam Score by Parental Involvement",
x = "Parental Involvement", y = "Mean Score") +
theme_minimal()
```

### *A.7 One-Way ANOVA Tests*

```
# Run one-way ANOVA for each categorical predictor
anova_results <- map_dfr(cat_var_names, function(var) {
formula <- as.formula(paste("exam_score ~", var))
model <- aov(formula, data = df_imputed)
tidy_mod <- tidy(model) |> filter(term != "Residuals")
# Eta-squared (effect size)
ss_total <- sum(tidy(model)$sumsq)
eta_sq <- tidy_mod$sumsq / ss_total
tibble(
variable = var,
f_stat = round(tidy_mod$statistic, 2),
p_value = round(tidy_mod$p.value, 4),
eta_sq = round(eta_sq, 4),
sig = tidy_mod$p.value < 0.05
)
})
# Display results sorted by F-statistic
anova_results |> arrange(desc(f_stat)) |> print(n = 20)
# Post-hoc Tukey HSD for significant factors (example: access_to_resources)
aov_model <- aov(exam_score ~ access_to_resources, data = df_imputed)
TukeyHSD(aov_model)
```

### *A.8 Multiple Regression Analysis*

```
# ■■ Data preparation: encode categorical variables ■■■■■■■■■■■■■■■■■■■■■■
df_model <- df_imputed |>
mutate(across(where(is.character), as.factor))
# ■■ Tidymodels workflow ■■■■■■■■■■■■■■■■■■■
```

```
tidy() |>
arrange(desc(abs(estimate))) |>
print(n = 25)
# ■■ VIF check for multicollinearity ■■■
```